

## BUS considerations

Most PC-based systems use the PCI (Peripheral Component Interconnect) bus to connect devices to the processor. The performance of the PCI is determined by the size of data path (bus width) and the speed at which it transfers the data (clock rate). Currently, the PCI supports the 32/64-bit bus width at a speed of 66 MHz.

There is also a parallel interface standard called SCSI (Small Computer System Interface). A typical problem that arises with the SCSI occurs when you place devices of different speeds on the same bus. So a device with a faster throughput may get 'stepped down' to the speed of the slower device on the bus. You can use the *sginfo* tool, part of the *sg3\_utils* package, to query or change the parameters of SCSI devices on your machine.

## Tuning the sequential read access

The kernel, when reading a file, always tries to take advantage of sequential disk access. 'Read-ahead' is based on the same assumption that if an application is accessing data from Block A, then the chances are more likely that the next Blocks—B, C and D will also need to be read. Therefore, by actually reading ahead and then caching the pages in memory, the kernel improves the I/O and reduces the response time. However, the read-ahead algorithm is designed to get turned off automatically whenever a random read request is detected.

The read-ahead function is based on two values:

1. The current window—the application reads pages from the buffer cache that is its current window.
2. The 'ahead' window—while the application reads pages from the current window, IO happens on the 'ahead' window.

When the application has finished reading from the current window, the 'ahead' window becomes the current window and a new 'ahead' window is started.

To view the current window size, issue *cat > /sys/block/sda/queue/read\_ahead\_kb*. The following is the command output for my machine:

```
# cat /sys/block/sda/queue/read_ahead_kb
128
```

As you can see, the current and default read-ahead is 128 KB. I can tune it to 256 KB, for example, as follows:

```
# echo 256 > /sys/block/sda/queue/read_ahead_kb
# cat /sys/block/sda/queue/read_ahead_kb
256
```

As is evident, the current window has now become 256 KB. However, note that rebooting the machine will change the default value to 128 KB. Please feel free to use the */etc/rc.local* file to make your changes permanent.

We can also use the *blockdev* command to report read-ahead in sectors. A sector is always 512 bytes in kernel 2.6.x.

```
# blockdev --report /dev/sda
```

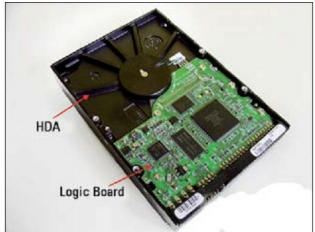


Figure 1: A typical hard disk drive

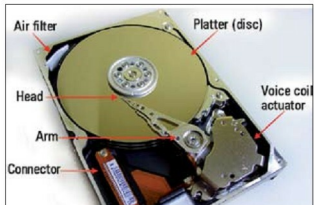


Figure 2: Components of the hard disk

```
RO RA SSZ BSZ StartSec Size Device
rw 256 512 4096 0 320072933376 /dev/sda
```

Remember that if page access occurs in the current window then the size of the new read-ahead windows increases by two pages. So because the default current window's size is 128 KB, the current read-ahead appears as 256. We can use the *blockdev* command with the following options to view and then set our read-ahead.

```
# blockdev --getra /dev/sda
256
# blockdev --setra 512 /dev/sda
# blockdev --getra /dev/sda
512
```

## Tuning the disk queue

The I/O subsystem is a series of processes with the responsibility to move blocks of data between the hard disk and the RAM. Generally, every computer task consists of a utility performing either one or both of the following:

- Read a block of data off the disk and move it to RAM
- Write a new block of data from the RAM to disk

These read/write requests are transformed into 'block device requests' that go into a queue. When adding an entry to